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Received: 20 December 2024 | Revised: 5 November 2025 | Accepted: 17 November 2025

DOI: 10.1002/ppp3.70153

Plants People Planet PPF Open Access

RESEARCH ARTICLE

Balancing tracks and trees: Assessing railroad impact on Brazilian biodiversity

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Funding information
São Paulo Research Foundation, Grant/Award Number: 2019/19544-7; National Council for Scientific and Technological Development, Grant/Award Number: 312389/2023-0; FAPESB, Grant/Award Number: PPF0013/2021; FAPESB postdoctoral fellowship, Grant/Award Number: BOL1132/2025

Societal Impact Statement

The Brazilian West–East Integration Railway (FIOL) aims to boost the national economy by improving commodity transport; however, it crosses three of Brazil's most biodiverse and fragile regions: the Caatinga, Cerrado and Atlantic Forest. Using digitised plant records and land-use analyses, our study reveals significant vegetation loss within the railway's area of influence between 2002 and 2021, particularly in the Cerrado. At least 13 species of high conservation value lost over 10% of their distribution area. These findings highlight the need to identify highly threatened species and promote sustainable practices that protect biodiversity while supporting economic development.

Summary

- The Brazilian West–East Integration Railway (FIOL) establishes a logistical transportation corridor to enhance the distribution of the country's commodities. Spanning 1527 km, it crosses transversally from the eastern coast to the interior of the country. Herbarium data, now widely accessible, have proven valuable for reconstructing the ecological history of specific regions. Here, we propose leveraging plant species records within the railway's area of influence, combined with species distribution modelling and temporal land-use change analysis, to assess which species of high conservation value have experienced loss of suitable habitat and to better understand the project's potential impacts on the flora.
- To accomplish this, we extracted herbarium records from GBIF using spatial queries within polygons covering the railway's area of influence to identify threatened species. We then developed distribution models for 104 species considered of high conservation value and integrated these with analyses of natural vegetation loss between 2002 and 2021.
- We identified 2217 native species within the railway's area of influence, of which 1000 are endemic and 120 are classified as threatened. Spatial analyses revealed significant native vegetation loss between 2002 and 2021, a period characterised

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by the expansion of agricultural areas alongside the beginning of railway construction.

- By integrating species occurrence data with vegetation loss, we found that 13 species of high conservation value lost more than 10% of their distribution area. The Cerrado emerged as a priority region, where railway construction poses a particularly high risk to the persistence of threatened species.

KEYWORDS
Brazilian Flora, Cerrado, environmental impact study, Ferrovia de Integração Leste Oeste, herbarium specimens

1 | INTRODUCTION

In the more than 500 years since Portuguese colonisation, Brazil's total population has expanded from an estimated 2 to 5 million Indigenous inhabitants to more than 200 million people, despite a drastic decline in its Indigenous population (IBGE, 1998, 2024a). This growth, initially concentrated mainly in the eastern regions close to the Atlantic coast, has expanded more intensively into central and western areas of the country in recent decades (Cabral, 2015). This expansion was primarily driven by the foundation of the nation's capital (Brasília) in the central region in 1960 and by the modernisation and growth of large-scale agriculture in the country's interior (Boaventura et al., 2023). Currently, Brazil ranks among the largest producers of commodities in the world. This status has come with significant environmental impacts in the central region of the country, including extensive deforestation (Lemes et al., 2019). One of the foremost challenges today is balancing economic growth with biodiversity conservation, particularly in Brazil, which harbours immense plant diversity (BFG, 2022) and where agriculture accounts for around 25% of Brazilian GDP (Buainain & Garcia, 2015; Newbold et al., 2015).

Transportation infrastructure, such as railways and roads, is pivotal in fostering regional integration, facilitating logistics, promoting social and cultural cohesion and driving development, contributing significantly to a country's overall growth (Nyumba et al., 2021). Interestingly, around 90% of new infrastructure projects have occurred in low- and middle-income countries, primarily in Africa, China and Latin America (Alamgir et al., 2017). The Brazilian West–East Integration Railway (FIOL, its Portuguese acronym) is a major Brazilian railway project, spanning a total length of 1527 km of track. It runs transversely across Brazil, starting at the export port on the coast in the southern region of Bahia state and extending to the southern region of Tocantins state (Figure 1). The primary objectives of this major project include transporting grains from a large commercial agriculture hub located in a new Brazilian agricultural frontier (Araújo et al., 2019) and transporting iron ore from southern Bahia to the port of Ilhéus, thereby facilitating the export of commodities (Infra, 2024). The construction of the railway began in 2011, and has proceeded in segments: Section I was built first, followed by Section II and finally Section III, which is still ongoing (Infra, 2024). It is important to highlight that Brazil's railway network project aims to connect the FIOL to the North–South Railway (FNS, its Portuguese acronym) in the city of Figueirópolis. Furthermore, with an even more ambitious goal, the FNS will be integrated with the Central-West Integration Railway (FICO, its Portuguese acronym), which will extend to the Peruvian border. This integration will give rise to the Transcontinental Railway, forming part of a linkage between the Atlantic Ocean, in Brazil, and the Pacific Ocean, in Peru (Infra, 2024).

It is well known that transportation infrastructure projects can cause numerous environmental impacts and risks, directly affecting the local ecosystem through construction or indirectly through increased human activities resulting from improved transit accessibility (Alamgir et al., 2017). Among the general negative impacts of a railway are atmospheric pollutant emissions and the loss of ecosystem integrity due to anthropogenic activities such as deforestation (Almeida et al., 2022; Cheng et al., 2020; Nyumba et al., 2021). Furthermore, this impact can be intensified when considering conservation units and areas of ecological interest (Alamgir et al., 2017; Bager et al., 2016). Due to the need for deforestation to build the right-of-way of the West–East Railway (Directly Affected Area [DAA]), as well as the parallel area for access to the tracks and the adjacent areas that may suffer several impacts (Area of Direct Influence [ADI]), it is essential to consider biotic environmental impacts, particularly those related to the flora. Alongside the perspective of biological diversity loss, the route of the FIOL passes through two global biodiversity hotspots: the Atlantic Forest and the Cerrado—the Brazilian savanna (Myers et al., 2000). These are areas listed among the 25 critical biodiversity hotspots on the planet, but, conversely, they experience exceptional habitat loss. In addition to these areas, the railway will also pass through the Caatinga domain—the largest and most biodiverse seasonally dry tropical forest (SDTF) region globally (Pennington et al., 2000).

Beyond studies in taxonomy and evolution, herbaria data have been valuable sources for reconstructing the history of a particular area or region (Greve et al., 2016; Williams & Crouch, 2017). The datasets obtained from these online repositories have been essential for documenting the impacts of global changes on botanical communities, such as predicting future impacts and aiding in the development of remediation strategies (James et al., 2018). Despite the known limitations of these databases and the caution required when working

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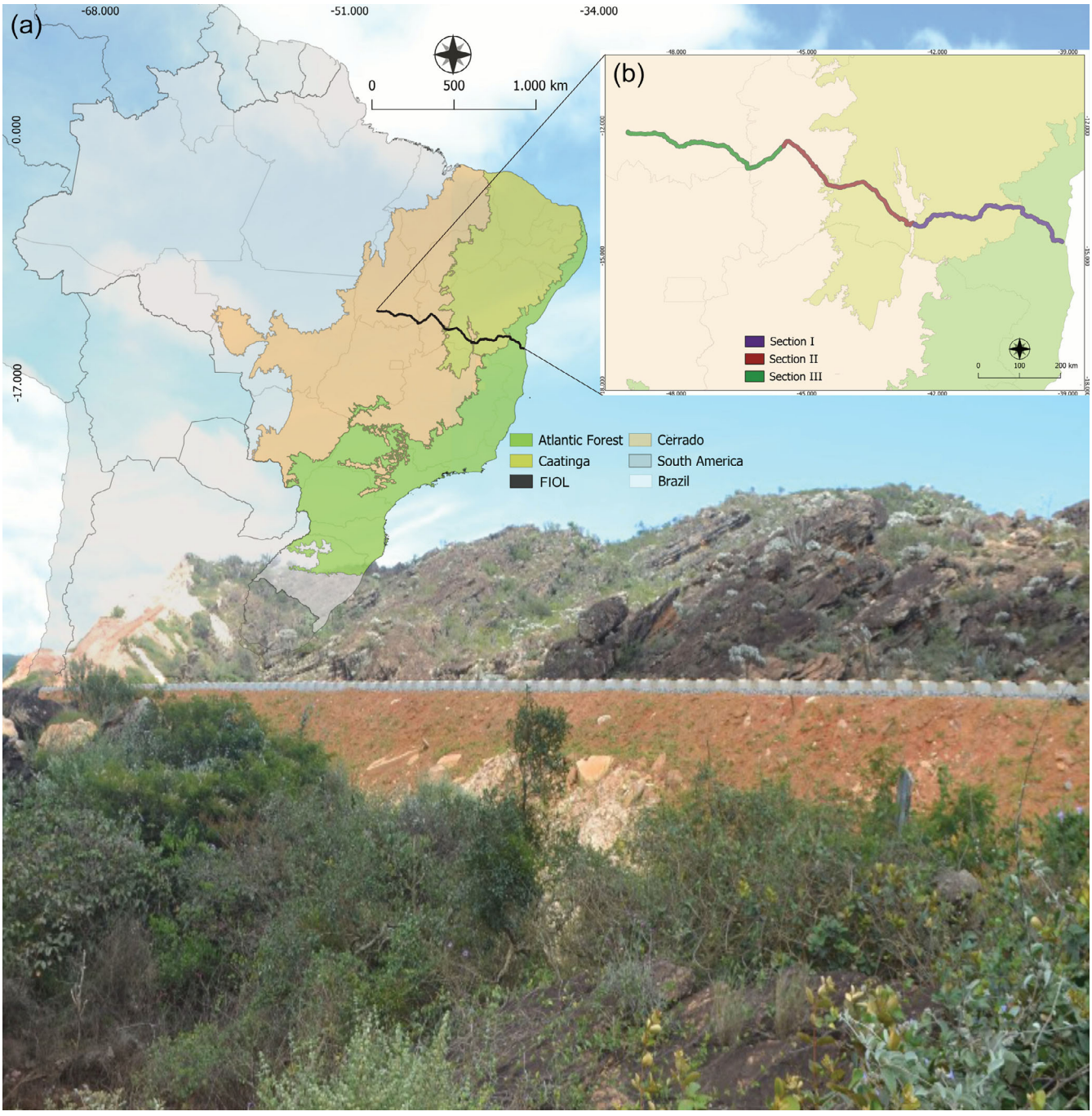


FIGURE 1 (a) The three phytogeographic domains of Brazil (Caatinga, Cerrado and Atlantic Forest) showing the entire route of the Brazilian West–East Integration Railway. (b) The railway route and the sections adopted in the land use spatial analyses. Landscape in the background of the railway track in 'Brejinho das Ametistas' in the municipality of Caetité (Bahia). Photo: Everton Hilo de Souza.

with them (Colli-Silva et al., 2020), the number of scientific papers published in recent years using herbarium data has increased significantly (Heberling & Isaac, 2017). In Brazil, for example, from 2009 to 2024, more than 8 million specimens have been digitised and made available online, distributed across 154 active herbaria in the country and centralised through two major projects: the INCT-HVFF, which makes its data available on SpeciesLink (SpeciesLink network, 2024), and Reflora project (Reflora, 2024). Both projects provide data to the Global Biodiversity Information Facility (GBIF.org, 2025). These current facilities for accessing large amounts of biodiversity data have played a key role in guiding conservation efforts in Brazil, providing increasingly robust information on species distributions in space and time, identifying knowledge gaps and supporting surveys of rare, endemic or threatened species (Pizzardo et al., 2024; Santana & Simon, 2022; Sousa-Baena et al., 2014a, 2014b; Versiane et al., 2025).

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Thus, considering the railway will pass through various areas of native vegetation and significant ecological interest, rare, endemic and/or endangered plant species may be affected. In this context, our objectives are (1) to survey georeferenced species collected in the area and stored in virtual herbaria; (2) to identify threatened and endemic species; (3) to analyse land-use change in the area of direct influence of FIOL through spatial analysis from 2002 to 2021; and (4) to integrate species occurrence data with vegetation loss analysis to assess the potential impact of railway construction on distributions of species with high conservation value.

2 | MATERIAL AND METHODS

2.1 | Study area

The delineation of our study area was established based on two criteria: the DAA, which will be deforested for the construction of the railway, and from there, the ADI, which corresponds to the DAA plus the parallel access area to the tracks, extending 5 km on each side. To delineate the study area, first, we used the railway shapefile, corresponding to the DAA, to map the entire FIOL route in QGIS version 3.34 (QGIS, 2024). The buffer algorithm was applied to determine the ADI, which calculated an area with a distance of 5 km from each feature of the railway route layer (totalling 10 km in width). Considering the length of the railway, with 1527 km, the ADI comprises 1,448,119 ha, spanning the states of Bahia and Tocantins and a small part of Goiás. It begins on the coast of Bahia (at the Port of Ilhéus) and ends in Figueirópolis, in the southeastern part of Tocantins (Figure 1). The railway was divided into three sections for construction: Section I, which starts in Ilhéus/BA and goes to Caetité/BA, covering 537 km; Section II, which runs from Caetité/BA to Barreiras/BA, with an extension of 485 km; and Section III, which extends from Barreiras/BA to Figueirópolis/TO, with an extension of 505 km (Figure 1b). The sections are related to the construction of FIOL: Section I corresponds to construction lots 1F, 02F, 02FA (Jequié Tunnel), 03F and 04F; Section II includes lots 05F, 05FA (bridge over the São Francisco River), 06F and 07F; and Section III consists of lots not yet defined, as it is still undergoing environmental licensing. Of these three sections, only the third has not begun construction and is currently under study (Infra, 2024). The railway also traverses the hydrographic basins of the Tocantins–Araguaia River, the São Francisco River and the Eastern Atlantic. Additionally, it crosses multiple biogeographical domains characterised by distinct vegetation types, including the Brazilian savanna (Cerrado), SDTF (Caatinga) and dense ombrophilous forest (Atlantic Forest) (IBGE, 2024b).

2.2 | Retrieving and verifying data on plant occurrence records from the FIOL area

To download the occurrences, it was necessary to divide the area into simple and precise polygons. Additionally, to obtain a more accurate checklist, we created the polygons based on the boundaries of the phytogeographic domains. Thus, the DIA was divided into 14 fragments (smaller polygons) using the Google Earth website (Google, 2024), with the Caatinga comprising seven polygons, Cerrado four and Atlantic Forest three. The polygons' files were saved in the KML format. Finally, the occurrences stored (only preserved species) on the GBIF platform (GBIF.org, 2025) were extracted using a function that incorporates the polygons in R software (R Core Team, 2024). The necessary packages for this function were 'rgbif' (Chamberlain et al., 2024), 'taxize' (Chamberlain, 2020), 'tidyverse' (Wickham et al., 2019) and 'sf' (Pebesma & Bivand, 2023). All the occurrence data and their respective DOI are available in Table S1.

The data cleaning and verification were carried out manually. The names of the collectors (marked in GBIF as 'recordedBy') and the associated collection number ('recordNumber') were standardised. In this way, duplicate samples among different herbaria were excluded. Subsequently, we classified each species as native, endemic, non-native and non-endemic by consulting the Flora and Funga of Brazil (2025). For endemic species verification, we have considered only the occurrences identified by specialists and checked individually in the respective taxonomic groups. In cases of uncertainty, some records were forwarded directly to specialist researchers for confirmation or clarification regarding the determination (see the Acknowledgements section). Records not identified to the species level, non-native species and those without identification confirmation were excluded. After this process, we identified the threatened species by consulting three Red Lists, the International Union for Conservation of Nature (IUCN, 2024), the Brazilian National Center of Flora Conservation (CNCFlora, 2025) and the available assessment of the threatened flora of the state of Bahia (SEMA, 2024). When a species is listed in multiple red lists in different categories, we prioritise the one in which it is classified as the most threatened.

2.3 | Retrieving and verifying data on species of high conservation value and species distribution modelling

Subsequently, considering only the threatened species identified in the previous step, a new retrieval of occurrence records for each of these species was carried out for species distribution modelling. The occurrences stored (only preserved species) on the GBIF platform (GBIF.org, 2025) were retrieved using the R packages 'rgbif' v.3.8.1 (Chamberlain & Boettiger, 2017; Chamberlain et al., 2024). The initial search returned 42,722 records for 120 species of angiosperms (Table S1). This dataset was processed using the 'plantR' v.0.1.8 pipeline (Lima et al., 2021) for taxonomic standardisation, geolocation validation and duplicate identification. In particular, collector names and collection numbers were harmonised to allow duplicate filtering across herbaria. Only records with valid coordinates and taxonomic identifications were retained. After cleaning, the dataset comprised 32,948 validated records for 118 species. The full list of all specimen occurrence records after data cleaning is available as Dataset S1.

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After data cleaning and filtering, the occurrence dataset contained 116 species and 10,011 unique georeferenced locations. Species distribution models (SDMs) were then constructed using this dataset; however, only species with more than three unique localities proceeded to the modelling stage. This reduced the pool to 110 species. Further exclusions were applied based on model evaluation metrics (see below), resulting in a final set of 104 species with satisfactory predictive performance used in subsequence analyses. Dataset S2 displays the list of unique locations used as the input for SDMs. SDMs were developed using the 'sdm' v.1.2.55 package in R (Naimi & Araújo, 2016). Environmental variables included topographic and climatic predictors preprocessed to a common extent and resolution, aligned to a 0.05° raster mask of Brazil. Collinearity among variables was assessed using variance inflation factor (VIF) analysis via the 'usdm' v.2.1.7 package (Naimi & Naimi, 2017), and only variables with VIF < 5 were retained for modelling. For each species, a minimum convex polygon was generated around its occurrences and buffered by 1000 km to define the calibration area. Background points were sampled from within this area, and models were trained using an ensemble of five algorithms: random forest (RF), support vector machine (SVM), multivariate adaptive regression splines (MARS), generalised linear model (GLM) and generalised additive model (GAM). Models were evaluated with fivefold cross-validation. Final ensemble predictions were thresholded using the mean of the sensitivity–specificity equality criterion, generating binary habitat suitability maps for each species. These binary rasters were saved and used in downstream analyses of habitat loss.

2.4 | Land cover and use of the ADI

To perform land cover and use analyses along the FIOL route, we used the databases from MapBiomas (MapBiomas, 2023). MapBiomas is a project created in 2015 in Brazil that conducts annual mapping of land cover and use and monitors water surface and fire scars. It operates collaboratively with NGOs, universities, laboratories and tech startups (Souza et al., 2020). The project's purpose is to map Brazil's land cover and land use and monitor the territory's changes (MapBiomas, 2023). To assess land cover changes along the railway's ADI, we used annual land use and land cover maps from the MapBiomas Project Collection 8.0. This dataset provides yearly classifications of 29 land-use types across Brazil from 2002 to 2021 at 30-m spatial resolution. Natural vegetation types were defined based on MapBiomas' class codes as follows: 3 (Forest Formation), 4 (Savanna Formation), 5 (Mangrove), 6 (Wetland), 11 (Natural Forest Regeneration), 12 (Other Natural Forest), 13 (Other Non-Forest Natural Formation), 49 (Floodplain Forest) and 50 (Restinga Forest). All other classes were grouped as anthropogenically impacted land. In QGIS version 3.22, the downloaded files were clipped to generate polygons representing the project's area of influence for each year. Subsequently, the 'r.report' tool was used to calculate the area of land use and land cover classes.

2.5 | Estimating potential plant distribution loss from land cover change around the railway project

To assess the impact of land cover change on the habitat of threatened and rare species, we overlaid annual land use and land cover rasters from the MapBiomas Project (MapBiomas, 2023) with binary distribution maps derived from SDMs. For each year from 2002 to 2021, we calculated the area of natural and impacted vegetation within the predicted distribution of each species, based on the reclassification of MapBiomas land cover classes described in Section 2.3. Two main metrics were derived for each species: (1) annual habitat loss, defined as the year-to-year decrease in natural vegetation within the species' modelled range, and (2) cumulative habitat loss, calculated as the reduction in natural vegetation from the baseline year of 2002. These were computed in both absolute terms (hectares lost) and as percentages relative to the area available at the beginning of each time step or to 2002.

To contextualise these losses, we also considered the total extent of each species' modelled distribution (i.e. the full area predicted as suitable by the SDM), as previously calculated. Based on this, we computed the proportion of cumulative habitat loss (as of 2021) occurring within the ADI relative to each species' overall modelled distribution. This provided a consistent measure of the relative impact across species with different range sizes. Accordingly, our estimates refer to the relative habitat areas potentially impacted by railway development within ADI and do not account for the species' full geographic distribution.

To assess patterns of potential distributional area loss among species under IUCN threat categories, we fitted linear models in two steps. First, we tested whether cumulative percentage loss differed among species from the three threat categories. We constructed a linear model with the species' maximum cumulative area loss (percentage) in the final year of analysis as the response variable and IUCN threat status (Critically Endangered [CR], Endangered [EN], Vulnerable [VU]) as a categorical predictor. Next, we evaluated whether distributional area loss increased over time with project progression by fitting a linear mixed-effects model, with cumulative percentage loss per year as the response variable, time (in years) as a fixed effect and species identity as a random variable to account for repeated measures. For both models, we confirmed the assumptions of normality and homoscedasticity through visual inspection of residual plots, and all assumptions were met.

All analyses were conducted in R (R Core Team, 2024) using custom scripts that integrate species binary maps with reclassified MapBiomas land cover data, calculate annual summaries and fit the specified linear models.

3 | RESULTS

3.1 | List of recorded and threatened species in the ADI of FIOL

The number of records extracted from the GBIF platform totalled 13,446. After the filtering, verification and data cleaning process, this

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number was reduced to 2217 native species, of which 1000 are endemic to Brazil, and 120 were identified as threatened in at least one Red List, 429 endemic species were not classified regarding their degree of threat, corresponding to 42.9% of all endemic species (Table 1 and Dataset S3). Although data extraction was performed for the Plantae kingdom, there were limited records of Algae and Bryophyta, many with erroneously inferred locations, and a predominance of Tracheophyta, with few lycophytes and ferns and most species being angiosperms.

There was also a significant difference in the number of records among domains, which consequently affected the total number of species, including threatened species, recorded in each. We retrieved 5808 records representing 1063 species, of which 448 are endemic and 70 threatened in the Atlantic Forest; 5544 records representing 1004 species, with 452 endemic and 41 threatened in the Caatinga; and 2087 records representing 553 species, including 190 endemic and 17 threatened in the Cerrado (Figure S1).

The 10 angiosperm families with the highest number of threatened species are Fabaceae (11), Myrtaceae (9), Annonaceae (8), Myrtaceae (8), Chrysobalanaceae (5), Euphorbiaceae (5), Malpighiaceae (5), Sapotaceae (5), Apocynaceae and Orchidaceae (5). The monilophytes Anemiaceae and Pteridaceae appear on the list of threatened species with one species each (Figure S1). On the endangered species list, 50.4% are tree species, followed by 26.4% shrubs, 14.0% herbaceous plants (including epiphytes), 5.8% subshrubs and 3.3% other growth forms such as lianas and palms. The most threatened tree species are from the Atlantic Forest, primarily belonging to the Fabaceae and Myrtaceae families. In contrast, most threatened Cactaceae species are from drier areas such as the Caatinga and Cerrado. CR species are highly restricted and locally endemic. A significant portion of the tree species in the EN and VU categories are traditionally exploited for their timber and hold economic value (see Section 4). Another noteworthy result was the concentration of species at specific points along the entire route within ecotone areas (where different vegetation types converge), as observed in the regions of Jequié and Caetité, where the Atlantic Forest transitions into the Caatinga and the Caatinga into the Cerrado, respectively (Figure 2). In Caetité, an additional significant factor is the presence of highland rock fields ('Serra do Brejinho das Ametistas'; Figure 1), which exhibit a high concentration of collections and species diversity.

3.2 | The impact of vegetation loss on the range of species with high conservation value

Our analyses indicate a maximum potential reduction of up to 30% in the species' distribution area within the zone most intensively impacted by the railroad over the past 10 years, with 13 plant species potentially losing 10% or more of their distribution area (Table 1 and Figure 3a). Of these, eight occur in the Cerrado, four

TABLE 1 Summary information on plant diversity compiled for the defined study area—Area of Direct Influence (ADI) of the Brazilian West-East Integration Railway (FIOL) (Dataset S3).

Data	Values	Species with >10% habitat loss (2002–2021)
Occurrence records obtained from GBIF (before data cleaning)	13,446	—
Occurrence species after data cleaning (elimination of records with doubtful taxonomic identification, non-native species, imprecise locality, duplicated specimens, unconfirmed occurrence)	2217	—
Number of species endemic to Brazil	1000	—
Number of species endemic not evaluated for conservation status	429	—
Number of species evaluated for conservation status by CNCFlora, IUCN and Red List of Bahia	120	—
Number of species for each list (species that appeared exclusively on a specific list)	CNCFlora 95 (40); IUCN 64 (23); Red List of Bahia 30 (1)	—
Number of species shared in all lists*; shared by CNCflora and IUCN*; shared by CNCflora and Red List of Bahia**; shared by IUCN and Red List of Bahia***	14*; 28**; 13***; 2****	—
Critically endangered	8	*Diplusodon gracilis*
Endangered	63	*Bunchosia pernambucana*, *Caryocar coriaceum*, *Cereus mirabella*, *Dyckia viridiflora*, *Handroanthus serratifolius*, *Hypolytrum bahiense*, *Kielmeyera occhioniana*, *Oryctina subaphylla*
Vulnerable	49	*Ficus calyptroceras*, *Manihot gracilis* (but see Section 4), *Melocactus salvadorensis*, *Mouriri gardneri*

Note: The asterisks indicate the number of species shared among the lists; dashes indicate when the information is not applicable.

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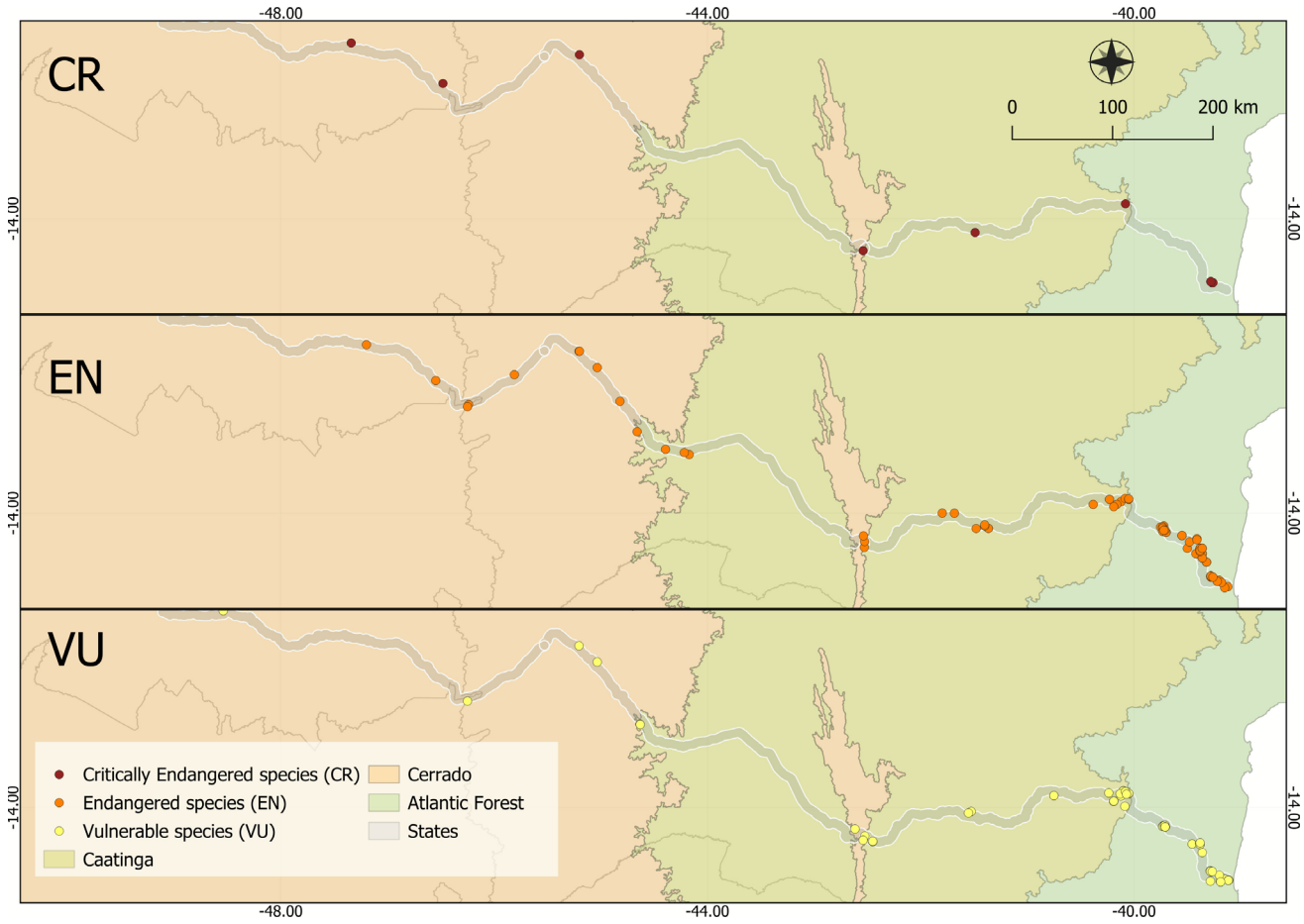


FIGURE 2 Map of threatened species collected within the railway's area of influence. Critically Endangered (CR) species are represented by red dots, Endangered (EN) by orange dots and Vulnerable (VU) by yellow dots. Certain areas discussed in the text, such as ecotone regions, are highlighted by a circle in Jequié and a triangle in Caetité ('Brejinho das Ametistas'). The star and diamond represent critically endangered species in Cerrado areas of São Desidério (Bahia) and Taguatinga (Tocantins), respectively.

in the Atlantic Forest and one in the Caatinga (Dataset S3). Notably, *Diplusodon gracilis* Koehne (Lythraceae), an endemic species of the Cerrado listed as CR by the IUCN, is among the most severely impacted. Regarding the total range loss of each species (not only within the FIOL corridor), 98% of the species lose less than 1% of their distribution area. However, three species stand out with substantially greater reductions: *Melocactus salvadorensis* Werderm. (3.1%), *Dyckia viridiflora* Forzza (7.14%) and *Hibiscus peterianus* Gürke (22.74%), with the latter two classified as EN on threatened species lists (Supporting Information S5). Comparison across the three IUCN threat categories revealed no differences in mean potential area loss, which consistently averaged 5% for each category (Figure 3b; $F_{2,74} = 0.7$; $p = .52$). Temporal trend analysis within the railway buffer zone showed that most species experienced increasing rates of potential habitat contraction over time, though the magnitude of these losses varied substantially among taxa (Figure 3c). Overall, the aggregate rate of potential distributional area loss for the threatened species was estimated at 1% per 3-year interval, as indicated by the black dashed regression line (Figure 3c; $F_{1.1462} = 2750.1$; $p < .001$).

4 | DISCUSSION

This study assesses the potential impacts of the FIOL railroad on plant species of high conservation value within its direct influence area. We compiled georeferenced species records from virtual herbaria, identifying 2217 native species, of which 1000 are endemic from Brazil and 120 are threatened within the railroad's area. Spatial analyses revealed significant native vegetation loss between 2002 and 2021, a period marked by both the agricultural expansion and the onset of railroad construction. By integrating species occurrence data and distribution models with vegetation loss patterns, we show that 13 species of high conservation value have lost over 10% of their distribution area, particularly in the last decade, coinciding with the onset of railway construction. However, this can be attributed not only to the construction itself but also to its location, particularly in the Cerrado (Brazilian savanna) portion, where there has been a recent expansion of agriculture for commodity production, which was one of the main reasons for building the railway. The Cerrado emerged as a priority region, where railway construction poses a particularly high risk to the

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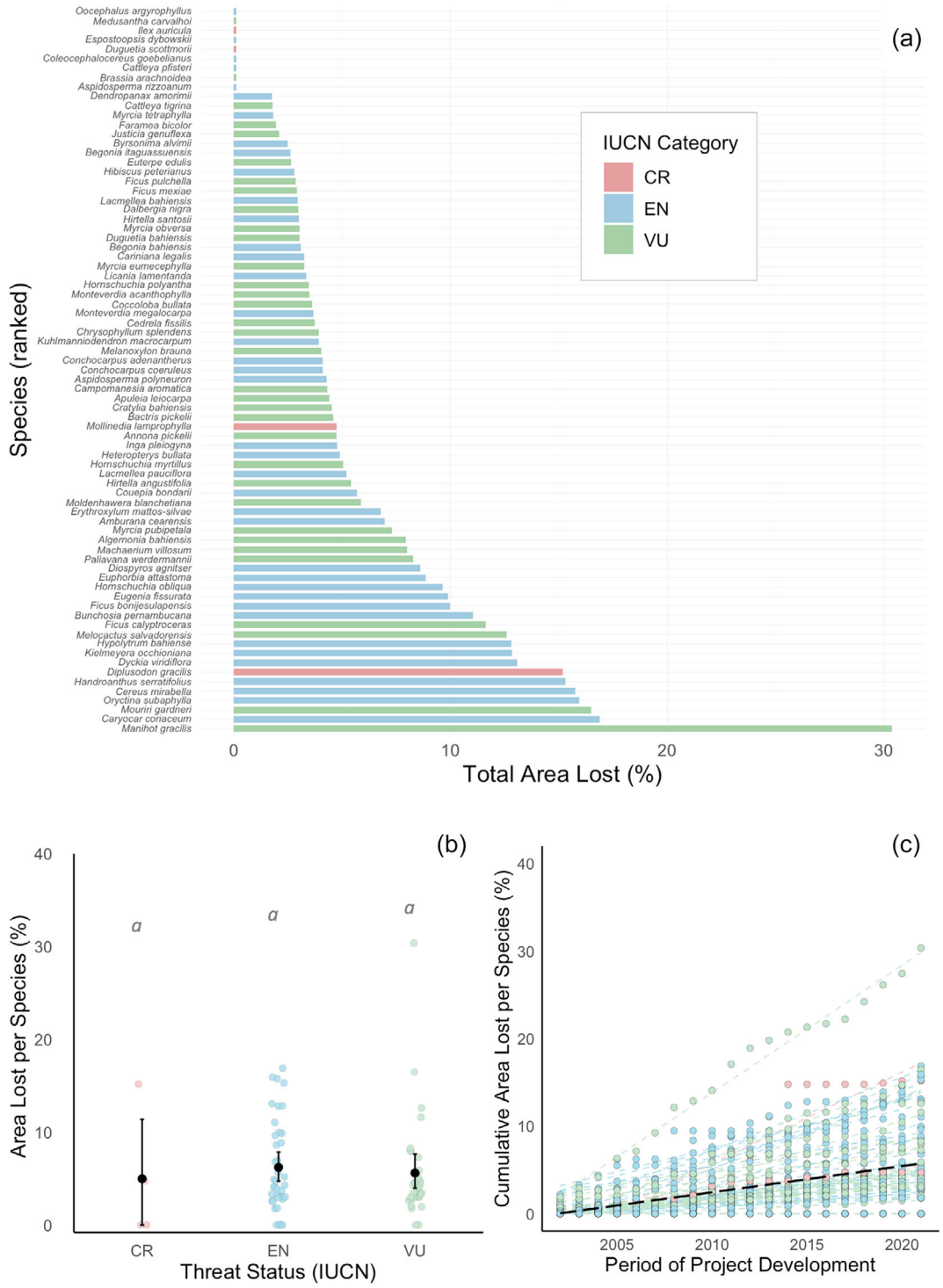


FIGURE 3 Potential area loss of IUCN-listed plant species in relation to the construction of the railway line. (a) Variation in potential area loss for each species within the buffer zone surrounding the railway. (b) Comparison of potential distributional area loss among the three IUCN threat categories—Critically Endangered, Endangered and Vulnerable. The black point indicates the mean loss for each category, and the surrounding bars show the 95% confidence interval estimated by bootstrap. (c) Cumulative potential area loss of species over the duration of railway implementation. The central dashed line represents the average trend across all threatened species, estimated from a linear mixed-effects model. Colours in all panels denote IUCN threat status for each plant species: Critically Endangered in red (CR), Endangered in blue (EN) and Vulnerable in green (VU).

persistence of threatened plant species. Our results represent an effort to maximise the use of occurrence data from biological collections to infer the potential threat posed by infrastructure projects—such as the railway examined here—on local biodiversity. This approach can be scaled to assess broader infrastructure developments, particularly in countries like Brazil, where expansion is

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necessary but must be accompanied by strategies to evaluate and mitigate biodiversity loss.

In Brazil, numerous studies utilising scientific botanical collection databases have demonstrated a pronounced concentration of collection efforts in regions near major urban centres (Colli-Silva et al., 2020; Sousa-Baena et al., 2014a). Areas with higher population density, which also host the oldest and most prominent research institutions, exhibit significant collecting activity. In contrast, more remote regions, such as the central and northern parts of the country, show limited or near-absence of records (Oliveira et al., 2017; Sousa-Baena et al., 2014a, 2014b). Our results reveal the same pattern, presenting a cross-sectional view of Brazil with a decline in the number of records from east to west. This trend is also directly reflected in the biodiversity catalogued across different domains, impacting the number of threatened species throughout the study area, which clearly decreases as we move further into the interior of the country. Although it is widely recognised that the Atlantic Forest domain along the country's coast is the most diverse and has the highest rates of endemism (BFG, 2022), it is also clear that it encompasses some of the best-studied and well-collected areas in Brazil (Colli-Silva et al., 2020). In our results, a high number of records were also observed to be concentrated in the Atlantic Forest. Consequently, Atlantic Forest has the highest numbers of records (6318), 45.3% off all data. Even after data cleaning, the records show the highest levels of diversity and threats in the FIOL section, with 1063 species. As one of the most fragmented and threatened domains in Brazil, it also has the highest number of evaluated threatened species (70 spp.).

Moving further inland, after leaving the Atlantic Forest domain in the coast, we observed a drastic reduction in records to the west. In the Cerrado domain, our results show a lower number of Brazilian endemic (190 spp.) and threatened species (17 spp.). Among the high conservation value species projected to lose the largest portions of their distribution ranges, those experiencing the greatest impacts on their occurrences are primarily associated with the sections of the railway crossing Cerrado domain, such as *Dyckia viridiflora* Forzza, *Cereus mirabella* N.P.Taylor, *Caryocar coriaceum* Wittm., *Hypolytrum bahiense* M. Alves & W.W.Thomas, *Manihot gracilis* Pohl (see further details below), *Oryctina subaphylla* Rizzini, *Mouriri gardneri* Triana and *Diplusodon gracilis*. The latter is a CR species (Oliveira & Messina, 2012), known from only a few collection records located precisely within the highway's area of influence in the Cerrado section. The low number of collections, combined with extensive land-use changes and substantial vegetation loss, highlights this region as both a priority for botanical surveys and a critical area for improving sampling efforts. Although *Hibiscus peterianus* shows less than 10% loss within the analysed corridor, when we consider its total distribution range, the loss becomes much more significant. This is because it is a narrowly distributed species, occurring mainly in the area surrounding the development (Fernandez & Gomes, R). Indeed, land-use changes in the Cerrado have yielded alarming deforestation rates in recent decades, making it the region with the highest current loss of native vegetation in Brazil, with less than 50% of its original extent (Alencar et al., 2020). From 2013 to 2023, the Cerrado lost 101.800 km^2 of native vegetation, 12% more than the Amazon (90.500 km^2), 71% more than the Caatinga (28.700 km^2) and 97% more than Atlantic Forest (2592.3 km^2) (INPE, 2024). In 2023 alone, the Cerrado accounted for 61% of all deforestation in Brazil compared to 25% in the Amazon (RAD 2023, 2024). With only 6.5% of its area under formal protection and increasing pressure from agriculture, the Cerrado is projected to lose an additional 31%–34% of its remaining vegetation by 2050 (Françoso et al., 2015; Strassburg et al., 2017). For the first time in the history of land use in Brazil, deforestation is now more prevalent in savanna formations than in forests (RAD 2023, 2024). Within the Cerrado, the MATOPIBA region, an agricultural frontier spanning Maranhão, Tocantins, Piauí and Bahia, has emerged as the region with the highest deforestation rates in the country (Araújo et al., 2019; RAD 2023, 2024; Santana & Simon, 2022). The railway under study is directly associated with transporting agricultural commodities from MATOPIBA to the port of Ilhéus. Our land-use analysis results encompass several MATOPIBA municipalities, such as São Desidério (the most deforested in 2023), and Barreiras (ranked tenth) (RAD, 2023; RAD, 2024). Although the Caatinga has also been heavily impacted by human activity (with 89% of its territory impacted; Araújo et al., 2023), our analyses showed few recent changes there, except for a notable increase in mining. Overall, the Cerrado exhibits the highest rates of vegetation loss and poses the greatest risk for reducing the distribution of threatened plant species along the railway route. The railway is not only a new source of direct impact but also accelerates fragmentation and landscape degradation.

The use of herbarium data for research has well-known restrictions and limitations that cannot be overlooked in the approach we implemented here. Besides geographic gaps, evidence that the Cerrado is under-sampled compared to other domains, we encountered additional challenges in analysing our data. These challenges are highlighted in other studies that used Digital Accessible Knowledge (Colli-Silva et al., 2020; Santana & Simon, 2022; Sousa-Baena et al., 2014a, 2014b), such as records excluded due to lack of identification at the species level, species not identified by specialists or cases where we could not confirm the identification. Undoubtedly, one of the major challenges in utilising Digital Accessible Knowledge to study species distributions or the flora of a specific region is the issue of geographic coordinates (Colli-Silva et al., 2020; Sousa-Baena et al., 2014b). Undoubtedly, one of the major challenges in utilising Digital Accessible Knowledge to study species distributions or the flora of a specific region concerns the quality and reliability of geographic coordinates. Many records lack coordinates entirely, while others present imprecise, outdated or erroneous georeferencing (e.g. swapped latitude/longitude, coordinates falling in the ocean or in administrative centroids), which can strongly compromise spatial analyses (Colli-Silva et al., 2020; Sousa-Baena et al., 2014b). Even when employing polygon delimitation to extract data, which theoretically should only include collections sampled within the area of interest, information from older collections without coordinates is often excluded, unless inferred from label data. However, these labels typically provide insufficient information for precise georeferencing. In some cases, even with accurate label data, coordinates

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may be erroneously inferred due to typographical or informational errors. Another point noted in our results was the large number of Brazilian endemic species that had not been assessed under threat categories (42.9%). Consequently, there are a substantial number of species with high conservation value in our dataset whose risk has not been evaluated. Conversely, this study includes species classified as vulnerable, which have been downlisted in more recent Brazilian assessments. This highlights that initial evaluations can, albeit less frequently, be based on incomplete data, largely dependent on the level of digitisation of botanical collection records. This is the case for the species *Manihot gracilis* and *Ficus calyptroceras* (Miq.) Miq. The former, classified here as Vulnerable, is listed as Least Concern (LC) in a more detailed assessment by Simon et al. (2020), which recognised it as having a significantly larger Extent of Occurrence and Area of Occupancy. This case, alongside that of *Ficus calyptroceras*, which has a more recent LC assessment not yet updated by the IUCN (Pederneiras et al., 2012), underscores the dynamic nature of species assessments. It demonstrates that even previously assessed species should be re-evaluated as the digitisation of botanical collections improves and more precise georeferencing techniques are applied to older, often scarce locality records. Collectively, these challenges magnify the potential impact of such projects, the lack of comprehensive local flora surveys and the consequences of natural habitat loss. Despite the inherent limitations of herbarium records, such as collection bias, and the constraints of IUCN classifications, our results demonstrate that this approach provides a robust initial estimate of risks to flora in contexts requiring rapid decision making.

The strategic location of the FIOL railway exemplifies Brazil's current land-use trajectory, where vegetation loss driven by the expansion of agriculture occurs in areas that remain poorly surveyed for plant diversity. This highlights the need to go beyond merely protecting species from land-use change; it calls for more integrated approaches that align biodiversity conservation with infrastructure development. Thus, the species lists generated in our study can support a range of conservation actions, including targeted species rescue, propagation research and evidence-based decision makers. Although Brazilian law mandates Environmental Impact Assessments (EIAs) before infrastructure projects can be licensed—and requires special attention when rare or threatened species are found (Ruaro et al., 2021), making EIAs a key mechanism for biodiversity protection—in practice, these surveys are often outsourced and conducted during limited, one-time field campaigns. As a result, they tend to reflect only a snapshot of local biodiversity, weakening the robustness of the data used to authorise construction. While our own dataset is not without gaps and limitations, it leverages historical and georeferenced herbarium data to provide a broader and more representative view of the regional flora than typical one-off floristic surveys restricted to a single sampling period. In this way, leveraging herbarium data and SDMs could complement and strengthen EIA processes. Herbarium records provide verifiable species occurrences that often capture a broader temporal and spatial context than short-term field surveys. SDMs, in turn, help identify suitable habitats and potential areas of conservation concern beyond the sampled sites. Incorporating these tools allows EIAs to better reflect real species distributions and landscape-level patterns, which is particularly important in large-scale developments.

Our approach, by maximising the use of biological collections and integrating land-use change analysis, combined with the usual field campaigns, may contribute to a scalable framework that can support biodiversity assessments across other rapidly low- and middle-income regions. We also envision future refinements to this framework, including the incorporation of predictive algorithms capable of identifying areas most likely to be impacted, while accounting for current knowledge gaps and species threat assessments.

AUTHOR CONTRIBUTIONS

Ana Luiza Silva Rocha: Conceptualisation of the study; data curation; formal analysis; figures; writing—original draft. **Juliana Gastaldello Rando:** Conceptualisation of the study; data curation; figures; writing—original draft; writing—review and editing. **Matheus Colli-Silva:** Data curation; formal analysis; writing—review and editing. **Anselmo Nogueira:** Formal analysis; figures; writing—original draft; writing—review and editing.

ACKNOWLEDGEMENTS

We extend our gratitude to the specialists who assisted us in confirming the specimens, namely, André Amorim (Malpighiaceae), Daniela Zappi (Cactaceae), José Rubens Pirani (Proteaceae and Rutaceae), Inês Cordeiro (Picrodendraceae), Jomar Jardim (Orchidaceae), Luiz Henrique Fonseca (Bignoniaceae), Marcelo Trovó (Eriocaulaceae), Daniela Zappi and Nigel Taylor (Cactaceae). We also extend our gratitude to the Graduate Program in Environmental Sciences (PPGCA-UFOB) and to the FAPESB project PPF0013/2021 for their support. This work forms part of the first author's master's dissertation developed within the PPGCA-UFOB. Juliana Gastaldello Rando acknowledges the FAPESB postdoctoral fellowship (BOL1132/2025). Anselmo Nogueira acknowledges support from the São Paulo Research Foundation (FAPESP) through a Young Investigator Grant (grant no. 2019/19544-7) and from the National Council for Scientific and Technological Development (CNPq) through a Research Productivity Fellowship (grant no. 312389/2023-0).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Most data are available in the Supporting Information, and all files related to the species distribution models, including raster files, ensembles, plots and statistics are available in the Figshare repository at https://figshare.com/projects/Balancing_Tracks_and_Trees_Assessing_Railroad_Impact_on_Brazilian_Biodiversity/251519. For further details, please contact the authors.

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How to cite this article: Rocha, A. L. S., Nogueira, A., Colli-Silva, M., & Rando, J. G. (2025). Balancing tracks and trees: Assessing railroad impact on Brazilian biodiversity. *Plants, People, Planet*, 1–12. https://doi.org/10.1002/ppp3.70153

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